

MATHEMATICS - III**Two Marks****Unit I****1. Define Analytic Function.**

2. Verify the function $\omega = \sin z$ is analytic or not.
3. An Analytic Function with constant real part is also constant.
4. An Analytic Function with constant modulus is also constant.
5. Test the analyticity of the function $f(z) = e^x(\cos y + i \sin y)$.
6. IF $f(z)$ and $\overline{f(z)}$ are analytic in a region R, Show that $f(z)$ is constant in that region
7. Show that the function $u = 2xy + 3y$ is harmonic
8. Show that the function $u = 3x^2y - y^3$ is harmonic
9. Show that $x^2 + iy^3$ is not analytic anywhere
10. IF $f(z) = (x - y)^2 + 2i(x + y)$ show that the C-R equations are satisfied along the curve $x - y = 1$.

Unit II

1. Find the image of the circle $|z - 1| = 1$ under the mapping $w = \frac{1}{z}$.
2. Find invariant points of transformation $w = \frac{1+z}{1-z}$.
3. Find the residue of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ at all poles.
4. Show that the transformation $\omega = \frac{z-i}{z+i}$ maps the real axis in the z -plane into unit circle in the ω -plane.
5. Define Isolated singularity
6. Define Taylor's series
7. State Laurent's series
8. Find the invariant points For $w = \frac{6-z}{z}$
9. Discuss the transformation $w = \sinh z$
10. Discuss the transformation $w = C + Z$

Unit III

1. Evaluate $\int_C \frac{dz}{z+4}$, where C is the circle $|z| = 2$
2. Evaluate $\int_C \frac{dz}{(z-2)^2}$ where C is the circle $|z| = 2$
3. Define Residue of a Multiple pole.
4. Define Residue for a simple pole
5. State Cauchy's Integral Formula
6. Find the Residue of $\frac{z}{(z+1)(z-2)}$
7. Evaluate $\int_C \frac{e^{-z}}{z+1} dz$ where c is the circle $|z| = 2$.
8. State Cauchy integral Theorem
9. Find the Residue of $\frac{7z-2}{z(z+1)(z-2)}$
10. State Cauchy Residue Theorem

Unit IV

1. State the Dirichlet's conditions *
2. Find Fourier Constant a_0 for $f(x) = (\pi - x)$ in $0 < x < 2\pi$
3. Find the Fourier Sine Series for $f(x) = e^x$ in $0 < x < \pi$.
4. Find Fourier constant a_0 for $f(x) = (\pi - x)$ in $0 < x < 2\pi$
5. IF $f(x)$ is an even Function in $(-\pi, \pi)$ Write down the expressions for Fourier coefficients <
6. Find a_0 in the Fourier series for $f(x) = x + x^2$ in $-\pi < x < \pi$ *
7. Find Half Range Sine series of $f(x) = x$ in $0 < x < l$
8. Explain Odd and Even functions x
9. Find Half Range cosine series of $f(x) = x$ in $0 < x < l$
10. Define Euler's Formula for Finding Fourier Coefficients

Unit V

1. What is called Harmonic Analysis?
2. Find the RMS value of $y = x^2$ in $-\pi < x < \pi$.
3. Write down the Complex Form of the Fourier series in $(0, 2l)$.
4. Find the Root mean square value of $f(x) = x$ in the interval $(0, l)$.
5. Write the complex Form of Fourier series For $f(x)$ defined in the interval $(c, c + 2l)$.
6. State Parseval's Theorem on Fourier coefficients.
7. Find the RMS value of $f(x) = x$ in $0 < x < l$.

Unit I

Eleven Marks

1. Derive the necessary condition For $f(z)$ is analytic
2. Derive the C-R in Polar Form and also prove that $f(z) = z^n$ is analytic.
3. Both the Real and Imaginary parts of any analytic Function satisfy Laplace's equation.
4. IF $w = u + iv$ is an analytic Function, then the curves of the Family $u(x, y) = C_1$ cut Orthogonally the curves of the Family $v(x, y) = C_2$ where C_1, C_2 are constants.
5. IF $f(z)$ is a regular Function of z , Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2$
6. IF $f(z)$ is a regular Function of z , Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)| = 0$
7. Find the analytic Function $f(z) = u + iv$ given that $v = \frac{2 \sin x \sinh y}{\cos 2x + \cosh 2y}$
8. IF $f(z) = u + iv$ is analytic & given that $u + v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$
9. Find the analytic Function whose imaginary part is given by $y + e^x \cos y$.
10. Find $f(z) = u + iv$ For the given real part is $\log(x^2 + y^2)$ also Find v

Unit II

1. Find the bilinear transformation which maps the points $Z_1 = \infty, Z_2 = i, Z_3 = 0$ into the points $w_1 = 0, w_2 = i, w_3 = \infty$.
2. Discuss the transformation $w = z^2$
3. Discuss the transformation $w = z + \frac{1}{z}$
4. Expand Laurent's series For $f(z) = \frac{7z-2}{z(z+1)(z-2)}$ in $1 < |z+1| < 3$
5. Obtain Taylor's series For the Function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ valid For $|z| < 3$
6. Discuss the transformation $w = \frac{1}{z}$ and also Find the image of the infinite strips (i) $\frac{1}{4} < y < \frac{1}{2}$ (ii) $0 < y < \frac{1}{2}$
7. Find the bilinear transformation which maps the points $Z_1 = \infty, Z_2 = 1, Z_3 = 0$ into the points $w_1 = 0, w_2 = i, w_3 = \infty$.
8. Discuss the transformation $w = \sin z$ and also Find $w = \cos z$
9. Obtain the Laurent's series expansion For $f(z) = \frac{(z+2)(z-2)}{(z+1)(z+4)}$ which are valid in (i) $1 < |z| < 4$ (ii) $|z| > 4$
10. Find the bilinear transformation which maps the points $Z_1 = 2, Z_2 = i, Z_3 = -2$ into the points $w_1 = 1, w_2 = i, w_3 = -1$

Unit III

1. State and Prove Cauchy's Integral Formula
2. Using Cauchy's residue theorem Find $f(z) = \frac{1}{(z^2+4)^2}$ where C is the circle $|z-i| = 2$
3. Evaluate $\int_C \frac{z+4}{z^2+2z+5} dz$ where C is the circle $|z+1+i| = 2$ using Cauchy's Integral Formula
4. State and Prove Cauchy's Integral Theorem
5. Evaluate $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)}$ $a > 0, b > 0$, by using Contour Integration
6. Using Contour Integration, evaluate $\int_0^{2\pi} \frac{d\theta}{1-2a\sin\theta+a^2}$ $0 < a < 1$
7. Using Cauchy's residue theorem Find $f(z) = \frac{1}{(z^2+4)^2}$ where C is the circle $|z-i| = 2$
8. Using Cauchy's integral Formula evaluate $\int_C \frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)(z-2)} dz$ where C is $|z| = 3$
9. Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta$ using contour integration
10. Evaluate $\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta}$ using contour integration

Unit IV

1. Obtain the Fourier series expansion For the Function $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in the interval $0 < x < 2\pi$ and hence deduce that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} = \frac{\pi^2}{6}$ *Handwritten note: 1st part is particular of 2nd part. Hence deduce that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{6}$*
2. Find the Fourier series of $f(x) = |\cos x|$ in $-\pi < x < \pi$
3. Find the Fourier series For the Function $f(x) = (l-x)^2$ in the interval $0 < x < 2l$ and hence Deduce that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} = \frac{\pi^2}{6}$
4. Explain $f(x) = x \sin x$ as a cosine series in $0 < x < \pi$
5. Obtain the sine series For $f(x) = x$ in the interval $0 < x < \pi$ and hence prove that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} = \frac{\pi^2}{8}$ by using Parseval's identity
6. Find the Fourier sine series For $y = x$ in $0 < x < l$. By using Parseval's identity Show that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$
7. Obtain the Fourier series expansion For the Function $f(x) = \sqrt{1-\cos x}$ in $0 \leq x \leq 2\pi$ and hence deduce that $\sum_{n=1}^{\infty} \frac{1}{4n^2-1} = \frac{1}{2}$
8. Expand $f(x) = x^2$ where $-\pi < x < \pi$ in a Fourier series of periodicity 2π . Hence show that
(i) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \propto \frac{\pi^2}{6}$, (ii) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \propto \frac{\pi^2}{12}$, (iii) $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \propto \frac{\pi^2}{8}$
9. Find the Fourier series For the Function $f(x) = 2x - x^2$ in the interval $0 < x < 3$ and $F(x+3) = F(x)$
10. Explain $F(x) = \sin x$ in $0 < x < \pi$ in Fourier Cosine series

Unit V

1. Find the sine series of $f(x) = x$ in $(0, \pi)$ and hence Show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$
2. Find the complex Form of the Fourier series $f(x) = e^x$, $-\pi < x < \pi$.
3. Find the Fourier series expansion For Function $y=F(x)$ which is defined in $(0, 2\pi)$ by means of the table given below. Find the series up to third harmonic.

x:	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y:	1	1.4	1.9	1.7	1.5	1.2	1

4. Find the sine series of $f(x) = x$ in $(0, l)$ and hence show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$.
5. Find the complex Form of the Fourier series $f(x) = e^x, -\pi < x < \pi$.
6. Find the Fourier series expansion For Function $y=f(x)$ which is defined in $(0, 2\pi)$ by means of the table given below. Find the series up to third harmonic.

x:	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$
y:	0	9.2	14.4	17.8	17.3	11.7

7. Find the complex Form of the Fourier series of $f(x) = \sin x$ in $(0, \pi)$
8. The Following table gives the variation of a periodic Function over the period T

x:	0	$\frac{T}{6}$	$\frac{T}{3}$	$\frac{T}{2}$	$\frac{2T}{3}$	$\frac{5T}{6}$	T
y:	1.98	1.3	1.05	1.3	-0.88	-0.25	1.98

Show that $f(\theta) = 0.75 + 0.37 \cos \theta + 1.004 \sin \theta$ where $\theta = \frac{2\pi x}{T}$ (OR)

9. Find the complex Form of the Fourier series $f(x) = e^{ax}, -\pi < x < \pi$ in the Form
- $$e^{ax} = \frac{\sin h a \pi}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n (a + in)}{a^2 + n^2} e^{inx}$$

10. Obtain the First three harmonic For the Function $y=f(x)$ given below

X:	0	1	2	3	4	5
Y:	4	8	15	7	6	2

(1)

1. When do you say a function $f(z)$ is analytic at a point $z=z_0$?
 A function $f(z)$ is said to be analytic at a point z_0 , if f is differentiable not only at z_0 but at every point of some neighbourhood of z_0 .

2. State the necessary conditions for $f(z)$ to be analytic in polar co-ordinates.

Let $f(z) = u + iv$ be the analytic function of $z(r, \theta)$ and possessing first order partial derivatives $u_r, u_\theta, v_r, v_\theta$ satisfying the relation $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ & $\frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$.

3. Obtain the critical points of the transformation

$$w = z + \frac{a^2}{z}$$

given $w = z + \frac{a^2}{z}$

$$\frac{dw}{dz} = 1 - \frac{a^2}{z^2} \rightarrow (1)$$

Condition for finding critical points is $\frac{dw}{dz} = 0$

$$(1) \Rightarrow 1 - \frac{a^2}{z^2} = 0$$

$$1 = \frac{a^2}{z^2} \Rightarrow z^2 = a^2$$

$$\boxed{z = \pm a}$$

$\therefore z = \pm a$ are the critical points.

4. Check whether $w = \frac{1}{z}$ is a bilinear transformation or not. Justify your answer.

W.K.T, The standard form of bilinear transformation

$$w = \frac{az+b}{cz+d} = \frac{1}{z}$$

By equating, $a = 0$

$$b = 1$$

$$c = 1$$

$$d = 0$$

$$ad - bc = (0)(0) - (1)(1) \\ = -1 (\neq 0)$$

$\therefore w = \frac{1}{z}$ is a bilinear transformation

5. State Cauchy's Integral Theorem.

If $f(z)$ is analytic and $f'(z)$ is continuous at all points within and on a simple closed curve C , then

$$\int_C f(z) dz = 0.$$

6. Find the formula for residue of $f(z)$ at $z = z_0$, a double pole

If $z = a$ is a double pole (or) pole of order 2 of $f(z)$,

then
$$[\text{Res } f(z)]_{z=a} = \lim_{z \rightarrow a} \frac{d}{dz} [(z-a)^2 f(z)].$$

7. Define a periodic function with period p .

A function $f(t)$ is said to be periodic with period T if $f(t+T) = f(t)$, for all t and $T > 0$. The least value of T is called as period of the function $f(t)$.

8. State Dirichlet's conditions for the existence of Fourier series (or) when does a function possess a Fourier series in terms of trigonometric terms?

A function $f(x)$ can be developed as Fourier series when the following conditions known as Dirichlet's conditions are said to be satisfied.

i) $f(x)$ is periodic, single valued and finite

ii) $f(x)$ has finite number of discontinuities in any one period

iii) $f(x)$ has the finite number of maxima and minima.

(2)

Write down the formula for complex form Fourier series of $f(x)$ in $(0, 2\pi)$

The complex form of Fourier series of $f(x)$ in $(0, 2\pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}, \text{ where } C_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx.$$

10. State Parseval's theorem on Fourier series of $f(x)$ in $(-l, l)$

The Parseval's theorem on Fourier series of $f(x)$ in the interval $(-l, l)$ is given by

$$\frac{1}{l} \int_{-l}^l [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Verify whether the function $f(z) = \bar{z}$ is analytic or not.

Let $f(z) = \bar{z}$
 $u + iv = x - iy$

Equating Real parts

$$u = x$$

$$u_x = 1$$

$$u_y = 0$$

Equating Imaginary part,

$$v = -y$$

$$v_y = -1$$

$$v_x = 0$$

$u_x \neq v_y \therefore$ Hence CR equations are not satisfied
 $\therefore f(z)$ is not analytic.

Show that $U = 2x - x^3 + 3xy^2$ is Harmonic

The condition for harmonic of U is $U_{xx} + U_{yy} = 0$.

Given $u = 2x - x^3 + 3xy^2$

$$u_x = 2 - 3x^2 + 3y^2 \quad \left| \quad u_y = 6xy \right.$$

$$u_{xx} = -6x \quad \left| \quad u_{yy} = 6x \right.$$

$$\therefore u_{xx} + u_{yy} = -6x + 6x = 0$$

$\therefore U$ is harmonic.

Find the map of the circle $|z|=3$ under the
 $w=2z$

Given $w=2z$ (Replace $w=u+iv$ & $z=x+iy$)

$$u+iv = 2(x+iy)$$

$$x+iy = \frac{u}{2} + i \frac{v}{2}$$

$$x = \frac{u}{2}; y = \frac{v}{2}$$

$$\text{Also given } |z|=3 \Rightarrow \sqrt{x^2+y^2} = 3$$

$$x^2+y^2 = 9$$

$$\left(\frac{u}{2}\right)^2 + \left(\frac{v}{2}\right)^2 = 9$$

$$u^2+v^2 = 36$$

$$\Rightarrow u^2+v^2 = (6)^2$$

$\therefore |z|=3$ maps to a circle in w -plane from z -plane with centre origin & radius 6 under transformation $w=2z$

Prove that the bilinear transformation has at most two fixed points.

The fixed points of the bilinear mapping

$w = \frac{az+b}{cz+d}$ is obtained by replacing $w=z$
 then we have $z = \frac{az+b}{cz+d}$

$$z(cz+d) = az+b$$

$$cz^2 + (d-a)z - b = 0 \rightarrow (1)$$

The roots of the eqn. are,

$$z = \frac{-(d-a) \pm \sqrt{(d-a)^2 - 4(c)(-b)}}{2c}$$

$$z = \frac{-(d-a) \pm \sqrt{(d-a)^2 + 4bc}}{2c}$$

(3)

Case (i): In eqn ①, if $c \neq 0$ and $(d-a)^2 - 4bc \neq 0$, then the two roots are different. Hence there are two fixed points.

Case (ii): In eqn ①, if $c \neq 0$ & $(d-a)^2 - 4bc = 0$, then the two roots are equal. Hence there is only one fixed point.

Case (iii): In eqn ①, if $c = 0$ then we have $(d-a)z - b = 0$

$\therefore$ one root is infinity and other root is $z = \frac{b}{d-a}$ provided $d \neq a$. Thus we have one fixed point is infinity and other one is finite fixed point $z = \frac{b}{d-a}$.

Case (iv): In eqn ①, if $c = 0$ & $d = a$, then infinity is the only one fixed point.

Evaluate $\int_C \frac{z}{c(z-1)(z-2)} dz$, where C is the circle $|z| = \frac{1}{2}$.

Since $z=1$ & $z=2$ are simple poles and they lie outside to the circle $C: |z| = \frac{1}{2}$.

By using Cauchy integral theorem $\int_C \frac{z}{c(z-1)(z-2)} dz = 0$.

Calculate the residue of $f(z) = \frac{e^{2z}}{(z+1)^2}$ at its pole.

Given $f(z) = \frac{e^{2z}}{(z+1)^2}$

Put $Dr=0$; $(z+1)^2=0$

$z = -1$ is pole of order 2

W.K.T $[\text{Res } f(z)]_{z=a} = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$

$[\text{Res } f(z)]_{z=-1} = \lim_{z \rightarrow -1} \frac{1}{1!} \frac{d}{dz} \left[(z+1)^2 \frac{e^{2z}}{(z+1)^2} \right]$

$= \lim_{z \rightarrow -1} \frac{d}{dz} e^{2z}$

$= \lim_{z \rightarrow -1} 2e^{2z} = 2e^{-2}$

Obtain the constant a_0 of the Fourier series for function $f(x) = x^2$, $-\pi < x < \pi$.

given $f(x) = x^2$, $-\pi < x < \pi$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{\pi^3}{3} - 0 \right] = \frac{2\pi^2}{3}$$

$$\boxed{a_0 = \frac{2\pi^2}{3}}$$

Define the RMS value of $f(x)$ over the interval (a, b)

The RMS value of $y = f(x)$ in the interval (a, b) is

$$\bar{y} = \sqrt{\frac{1}{b-a} \int_a^b [f(x)]^2 dx}$$

$$\bar{y}^2 = \frac{1}{b-a} \int_a^b [f(x)]^2 dx \quad \text{--- (or) ---}$$

What is meant by Harmonic analysis?

The process of finding the Fourier series for a function given by numerical values is known as Harmonic Analysis.

Test the analyticity for $f(z) = z^n$ ④

Let $f(z) = z^n$

$$z = re^{i\theta}$$

$$f(z) = z^n = (re^{i\theta})^n = r^n e^{in\theta}$$

$$= r^n (\cos\theta + i \sin\theta)^n$$

$$u + iV = r^n (\cos n\theta + i \sin n\theta)$$

Equating real & imaginary parts on both sides,

$$u = r^n \cos n\theta \quad ; \quad V = r^n \sin n\theta$$

$$u_r = nr^{n-1} \cos n\theta \quad ; \quad V_r = nr^{n-1} \sin n\theta$$

$$u_\theta = -nr^{n-1} \sin n\theta \quad ; \quad V_\theta = nr^{n-1} \cos n\theta$$

W.K.T, C-R equations in polar form are

$$V_r = \frac{1}{r} V_\theta \quad \& \quad V_r = -\frac{1}{r} u_\theta$$

$$u_r = \frac{nr^{n-1} \cos n\theta}{r} \quad ; \quad V_r = \frac{nr^{n-1} \sin n\theta}{r}$$

$$u_r = nr^{n-1} \cos n\theta \quad ; \quad V_r = nr^{n-1} \sin n\theta$$

$$nr^{n-1} \cos n\theta = nr^{n-1} \cos n\theta \quad ; \quad nr^{n-1} \sin n\theta = nr^{n-1} \sin n\theta$$

$\therefore$ C-R equations are satisfied.

$\therefore$ Hence $f(z)$ is analytic.

Prove that $V = 3x^2y + x^2 - y^3 - y^2$ is harmonic

W.K.T, the conditions for harmonic, $V_{xx} + V_{yy} = 0$

given $V = 3x^2y + x^2 - y^3 - y^2$

$$\begin{array}{l|l} V_x = 6xy + 2x & V_y = 3x^2 - 3y^2 - 2y \\ V_{xx} = 6y + 2 & V_{yy} = -6y - 2 \end{array}$$

$$V_{xx} + V_{yy} = 6y + 2 - 6y - 2 = 0$$

$\therefore V$ is harmonic.

Find the invariant points of $w = \frac{1+z}{1-z}$

To find invariant points, Replace $w = z$

$$z = \frac{1+z}{1-z} \Rightarrow z - z^2 = 1 + z$$

$$z^2 = -1$$

$$z = \pm i$$

$z = \pm i$ are fixed points.

Define essential singularity with an example.

If the principal part contains infinite number of terms containing negative powers of $(z-a)$ i.e. all b_n 's are non-zero then $z=a$ is known as an essential singularity of $f(z)$.

eg: $f(z) = e^{1/z} = 1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$

where $z=0$ is an essential singularity, because $f(z)$ has an infinite series of negative powers of z .

Evaluate $\int_C \frac{dz}{z-2}$ where $C: |z|=3$.

Put $0 < \epsilon < 1$, $z-2=0$

$$z=2 \in C$$

$$\int_C \frac{dz}{z-2} = 2\pi i f(2)$$

$$= 2\pi i \quad [\because f(z) = 1 \text{ \& } z=2 \in C]$$

Find residue of $f(z) = \frac{z^2}{(z-1)(z-2)^2}$ at its poles.

given

$$f(z) = \frac{z^2}{(z-1)(z-2)^2}$$

⊗ $z=1$ is a pole of order 1

⊗ $z=2$ is a pole of order 2

$$\begin{aligned} \text{[Res } f(z)]_{z=1} &= \lim_{z \rightarrow 1} (z-1) \frac{z^2}{(z-1)(z-2)^2} \\ &= \lim_{z \rightarrow 1} \frac{z^2}{(z-2)^2} \end{aligned}$$

$$\begin{aligned} \text{[Res } f(z)]_{z=2} &= \lim_{z \rightarrow 2} \frac{1}{1!} \frac{d}{dz} [(z-2)^2 f(z)] \\ &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{(z-2)^2 z^2}{(z-1)(z-2)^2} \right] \\ &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{z^2}{(z-1)} \right] \\ &= \lim_{z \rightarrow 2} \left[\frac{(z-1)(2z) - z^2}{(z-1)^2} \right] \\ &= 0 \end{aligned}$$

If $f(x) = x^2 + x$ is expressed as a fourier series in the interval $(-2, 2)$ to which value this series converges $x=2$?
 Since $x=2$ is point of discontinuity.

$$\therefore f(x) = \frac{f(-2) + f(2)}{2} = \frac{(4-2) + (4+2)}{2} = 4$$

If the fourier series corresponding to $f(x) = x$ in the interval $(0, 2\pi)$ is $\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$.
 without finding the values of a_0, a_n & b_n find the value of $\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$.

$$\begin{aligned} \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) &= 2 \cdot \frac{1}{2\pi} \int_0^{2\pi} x^2 dx \\ &= \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{2\pi} \\ &= \frac{8\pi^2}{3} \end{aligned}$$

Write down the complex form of fourier series of $f(x)$ in $(0, 2l)$ and the Euler's formula for the above Fourier coefficient?

The complex form of Fourier series for $f(x)$ defined in the interval $(0, 2l)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{\frac{i n \pi x}{l}}$$

$$\text{where } C_n = \frac{1}{2l} \int_0^{2l} f(x) e^{-\frac{i n \pi x}{l}} dx.$$

Find the bilinear map which maps the points $\infty, i, 0$ of the plane onto $0, i, \infty$ of the plane.

~~then~~ $z_1 = \infty, z_2 = i, z_3 = 0; w_1 = 0, w_2 = i, w_3 = \infty$

The bilinear transformation is given by,

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-w_1)w_3\left(\frac{w_2}{w_3}-1\right)}{(w_1-w_2)w_3\left(1-\frac{w}{w_3}\right)} = \frac{z_1\left(\frac{z}{z_1}-1\right)(z_2-z_3)}{z_1\left(1-\frac{z_2}{z_1}\right)(z_3-z)}$$

$$\frac{(w-w_1)\left(\frac{w_2}{w_3}-1\right)}{(w_1-w_2)\left(1-\frac{w}{w_3}\right)} = \frac{\left(\frac{z}{z_1}-1\right)(z_2-z_3)}{\left(1-\frac{z_2}{z_1}\right)(z_3-z)}$$

$$\frac{(w-0)(0-1)}{(0-i)(1-0)} = \frac{(0-1)(i-0)}{(1-0)(0-z)}$$

$$\frac{w}{i} = \frac{i}{z} \Rightarrow w = -\frac{1}{z}$$

Find the fixed points of the mapping $w = \frac{3-z}{1+z}$.

For finding invariant points, replace w by z in the given transformation $w = \frac{3-z}{1+z}$

$$z = \frac{3-z}{1+z}$$

$$z + z^2 = 3 - z$$

$$z^2 + 2z - 3 = 0$$

$$(z-1)(z+3) = 0$$

$z = 1, -3$ are fixed points.

Find the residue of $f(z) = \frac{z}{(z-1)^2}$ at its poles.

Since $z=1$ is a pole of order 2 in $f(z)$

$$\begin{aligned} [\text{Res } f(z)]_{z=1} &= \lim_{z \rightarrow 1} \frac{1}{1!} \frac{d}{dz} \left[(z-1)^2 \frac{z}{(z-1)^2} \right] \\ &= \lim_{z \rightarrow 1} 1 = 1. \end{aligned}$$

Find b_n in $f(x) = x^2$ as a fourier series in $(-\pi, \pi)$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx \, dx$$

$$= 0 \quad [\because x^2 \sin nx \text{ is odd in interval}]$$

Define RMS value of $f(x) = x$ in the interval $(0, l)$

$$Y^2 = \frac{1}{l} \int_0^l [f(x)]^2 dx$$

$$= \frac{1}{l} \int_0^l x^2 dx = \frac{1}{l} \left[\frac{x^3}{3} \right]_0^l = \frac{l^2}{3}$$

$$Y = l/\sqrt{3}$$

1. Define continuity of a function of a complex variable.
A function $f(z)$ is said to be continuous at a point z_0 if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$. Simply say that, a function is said to be continuous in a domain if it is continuous at every point of the domain.

Write down the necessary and sufficient condition for Cauchy-Riemann equation to be analytic.

(i) Necessary condition

If $f(z) = u(x, y) + iv(x, y)$ is an analytic function of z , then the first order partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ should exist and satisfy

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} ; \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$(i.e) \quad U_x = V_y ; U_y = -V_x$$

(ii) Sufficient condition

Let $f(z) = u(x, y) + iv(x, y)$ be a single valued and continuous function possessing the four partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ at each point in the region R satisfying the C-R equations then $f(z)$ is analytic in the region R .

Find the map of the circle $|z| = k$ by the transformation $w = (1+i)z + 3+2i$

(7)
In the bilinear transformation $w = \frac{az+b}{cx+d}$, what is the necessity for the condition $ad-bc \neq 0$?

The condition $ad-bc \neq 0$ ensures that the bilinear transformation is conformal. If $ad-bc = 0$, then $f(z)$ would be identically constant. i.e. every point of the z -plane is critical point of the transformation.

State Cauchy Residue Theorem.

If $f(z)$ is analytic inside a simple closed curve C , except at a finite number of singularities $a_1, a_2, a_3, \dots, a_n$ lying inside C and if $R_1, R_2, R_3, \dots, R_n$ are the residues at these singularities then,

$$\oint_C f(z) dz = 2\pi i (\text{Sum of Residues}) = 2\pi i (R_1 + R_2 + \dots + R_n)$$

Explain what is meant by contour integration?

The evaluation of certain types of real definite integrals can be done by expressing them in terms of integrals of complex function over suitable closed paths or contours and applying Cauchy's Residues theorem. This process of evaluation of definite integrals is known as contour integration.

Define Fourier Series.

If $f(x)$ is a periodic function and satisfies Dirichlet's conditions, then it can be represented by an infinite series called Fourier Series.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where a_0, a_n, b_n ($n=1, 2, 3, \dots$) are called Fourier Coefficients of $f(x)$ and it is called as the

Trigonometric form of fourier series

Define Half Range series

It is required to expand a function $f(x)$ in the range $(0, \pi)$ or in the range $(0, l)$, then we represent $f(x)$ as cosine or sine series. Such series called as half range series.

Define Root mean square value

The root mean square value of the function $y = f(x)$ over an interval (a, b) is defined as

$$\bar{y}^2 = \frac{1}{b-a} \int_a^b y^2 dx$$

$$\bar{y} = \sqrt{\frac{1}{b-a} \int_a^b y^2 dx} \quad (\because y = f(x))$$

State Parseval's Identity theorem.

If the fourier series for $f(x)$ converges uniformly in $(-l, l)$ then

$$\bar{y}^2 = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

$$\text{where } \bar{y}^2 = \frac{1}{l} \int_{-l}^l y^2 dx$$

Examine whether the function $z^3 + z$ is analytic or not.

$$f(z) = z^3 + z$$

$$u + iv = (x + iy)^3 + (x + iy)$$

$$= x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 + x + iy$$

$$= x^3 + i3x^2y - 3xy^2 - iy^3 + x + iy$$

$$u + iv = (x^3 - 3xy^2 + x) + i(3x^2y - y^3 + y)$$

(8)

$$u = x^3 - 3xy^2 + x$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 1$$

$$\frac{\partial u}{\partial y} = -6xy$$

$$v = 3x^2y - y^3 + y$$

$$\frac{\partial v}{\partial x} = 6xy$$

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 + 1$$

By C-R eqn,

$$(*) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$3x^2 - 3y^2 + 1 = 3x^2 - 3y^2 + 1$$

$$(*) \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$-6xy = -6xy \Rightarrow 6xy = 6xy$$

$\therefore f(z)$ satisfy CR eqn

$\therefore f(z)$ is analytic.

State polar form of Cauchy - Riemann equation

Let (r, θ) be the polar co-ordinates of a point whose cartesian co-ordinates are (x, y) then the relation is $x = r \cos \theta$, $y = r \sin \theta$ and $z = x + iy = r(\cos \theta + i \sin \theta) = r e^{i\theta}$.

Find the image of $|z+1|=1$ under the mapping $w = \frac{1}{z}$

Given

$$|z+1|=1$$

$$|x+iy+1|=1 \Rightarrow |(x+1)+iy|=1$$

$$\sqrt{(x+1)^2 + y^2} = 1$$

$$(x+1)^2 + y^2 = 1$$

$$x^2 + 2x + 1 + y^2 = 1$$

$$x^2 + y^2 + 2x = 0 \rightarrow \textcircled{1}$$

Also, given $w = \frac{1}{z}$

$$u+iv = \frac{1}{x+iy} = \frac{u-iv}{u^2+v^2} = \frac{u}{u^2+v^2} - \frac{iv}{u^2+v^2}$$

$$x+iy = \frac{1}{u+iv}$$

$$x = \frac{u}{u^2+v^2}, \quad y = \frac{-v}{u^2+v^2} \rightarrow (2)$$

sub (2) in (1)

$$\left(\frac{u}{u^2+v^2}\right)^2 + \left(\frac{-v}{u^2+v^2}\right)^2 + 2\left(\frac{u}{u^2+v^2}\right) = 0$$

$$u^2+v^2 + 2u(u^2+v^2) = 0$$

$$[u^2+v^2](1+2u) = 0$$

$$1+2u = 0 \quad [\because u^2+v^2 \neq 0]$$

$\therefore 1+2u=0$ which is a eqn. of st. line in w -plane.

Find fixed points of transformation $w = \frac{1+z}{1+\bar{z}}$
To find fixed points, replace $w \rightarrow z$

$$z = \frac{1+z}{1+\bar{z}}$$

$$z(1+\bar{z}) = 1+z$$

$$z - z^2 = 1+z$$

$$z^2 = -1$$

$z = \pm i$ are the fixed points.

Evaluate $\int_C \frac{e^z}{z-1} dz$, where C is circle $|z|=2$.

Soln:

$$\text{given } f(z) = \frac{e^z}{z-1}$$

$$\text{Put } D_r = 0$$

$$z = 1$$

also given, $|z|=2$, $|1|=1 < 2$

$\therefore z=1 \in C$ (9)
Using Cauchy's Integral formula,

$$\begin{aligned} \int_C \frac{f(z)}{z-a} dz &= 2\pi i f(a) \\ &= 2\pi i f(1) \quad [\because f(z) = e^z] \\ &= 2\pi i [2.718] \quad [f(1) = e^1 = 2.718] \\ &= 5.436 \pi i \end{aligned}$$

If $f(x) = |x|$ in $-\pi < x < \pi$, find a_0 .

Given $f(x) = |x|$

W.K.T $|f(x)|$ is always even
 $\therefore$ It is an even function.

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x dx \quad \left[\because \int_{-\pi}^{\pi} = 2 \int_0^{\pi} \text{ (for even function)} \right] \\ &= \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \\ &= \frac{\pi^2}{\pi} = \pi \\ \boxed{a_0 = \pi} \end{aligned}$$

Define Harmonic Function
Any function, say $u(x, y)$ which possesses continuous second order partial derivatives and satisfies the Laplace equation $U_{xx} + U_{yy} = 0$, then it is called as harmonic function.

Test whether the function $f(z) = x^2 + iy^2$ is an analytic or not.

$$f(z) = x^2 + iy^2$$

$$u + iv = x^2 + iy^2$$

$$u = x^2$$

$$v = y^2$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial y} = 2y$$

By C-R eqn,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$; \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$0 = 0$$

$$2x \neq 2y$$

$\therefore$ Since $f(z)$ does not satisfy C-R equation $f(z)$ is not analytic.

Define Bilinear Transformation.

The transformation $w = \frac{az+b}{cx+d}$, where $ad-bc \neq 0$, and a, b, c & d being complex constants is called the bilinear transformation. It is also called as Möbius (or) Linear fractional transformation.

What are the types of singularities?

- i) Isolated singularity
- ii) Poles
- iii) Essential singularity
- iv) Removable singularity

(10)

evaluate, using Cauchy's integral formula $\int_C \frac{z}{z-2} dz$
 where C is the circle $|z|=1$

Given

$$\int_C \frac{z}{z-2} dz$$

$$C: |z|=1$$

Put $Dr=0$, $z-2=0$
 $z=2$

Also given, $|z|=1$
 $|2|=2 > 1$
 $\therefore z=2 \notin C$

$$\therefore \int_C \frac{z}{z-2} dz = 0.$$

Define even function with an example.

A function $f(x)$ is said to be an even function if $f(-x) = f(x)$.

example.

$$f(x) = x^2$$

Put $x = -x$

$$f(-x) = (-x)^2$$

$$= x^2$$

$$f(-x) = f(x)$$

$\therefore$ Hence it is an even function

Write C-R equations in Cartesian form

C-R equation

$$i) \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$ii) \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Show that the function $x^4 - 6x^2y^2 + y^4$ is harmonic.

Let $u = x^4 - 6x^2y^2 + y^4$

$$\frac{\partial u}{\partial x} = 4x^3 - 12xy^2$$

$$\frac{\partial^2 u}{\partial x^2} = 12x^2 - 12y^2$$

$$\frac{\partial u}{\partial y} = -12x^2y + 4y^3$$

$$\frac{\partial^2 u}{\partial y^2} = -12x^2 + 12y^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 12x^2 - 12y^2 - 12x^2 + 12y^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad [\because \text{Laplace eqn. satisfied}]$$

$\therefore$ The given function satisfies Laplace eqn.
 $\therefore$ The given function is harmonic.

Find the image of the circle $|z| = 2$ by the transformation
 $w = z + 3 + 2i$

Soln :

given $|z| = 2$

$$|x + iy| = 2$$

$$\sqrt{x^2 + y^2} = 2$$

$$x^2 + y^2 = 4 \rightarrow (1)$$

Also, $w = z + 3 + 2i$

$$u + iv = x + iy + 3 + 2i$$

$$u + iv - 3 - 2i = x + iy$$

$$(u - 3) + i(v - 2) = x + iy$$

Equating Real & imaginary parts,

$$x = u - 3 \rightarrow (3); \quad y = v - 2 \rightarrow (2)$$

② & ③ in ①

$$(u-3)^2 + (v-2)^2 = 4$$

$$u^2 - 6u + 9 + v^2 - 4v + 4 = 4$$

$$u^2 - 6u + v^2 - 4v + 9 = 0$$

circle in w -plane.

State Taylor's theorem:

If $f(z)$ is analytic inside a circle C with centre at z_0 and radius r , then at each point z inside C ,

$$f(z) = f(z_0) + \frac{(z-z_0)}{1!} f'(z_0) + \frac{(z-z_0)^2}{2!} f''(z_0) + \dots$$

This series is called as Taylor's series for the function $f(z)$ at/about $z = z_0$. The above Taylor series can be represented as

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n \text{ where } a_n = \frac{f^{(n)}(z_0)}{n!}$$

Evaluate $\int_C \frac{dz}{2z+1}$ where C is $|z|=1$

$$\text{given } f(z) = \frac{1}{2z+1}$$

$$\text{Put } 2z+1=0$$

$$2z+1=0$$

$$z = -\frac{1}{2}$$

Also given, $|z|=1$

$$|-\frac{1}{2}| = \frac{1}{2} < 1$$

$$\therefore z = -\frac{1}{2} \in C$$

Using Cauchy's integral formula,

$$\int_C \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

$$\int_C \frac{dz}{2z+1} = 2\pi i f\left(-\frac{1}{2}\right)$$

$$= 2\pi i (1) \quad [\because f(z)=1, f(-\frac{1}{2})=1]$$

$$\int_C \frac{dz}{2z+1} = 2\pi i$$

Find the constant a_0 of the Fourier series for the function $f(x) = x$ in $0 \leq x \leq 2\pi$

Soln:

given $f(x) = x$, $0 \leq x \leq 2\pi$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} [4\pi^2]$$

$$\boxed{a_0 = 2\pi}$$

State Parseval's identity in $(0, 2l)$.

If the Fourier series for $f(x)$ converges uniformly in $(0, 2l)$ then $\overline{y^2} = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$

where $\overline{y^2} = \frac{1}{l} \int_0^l y^2 dx \quad [\because y = f(x)]$

the root mean square ^(R.M.S) value of x^2 in $(0, \pi)$.
given $f(x) = x^2$ in $(0, \pi)$

$$\bar{y}^2 = \frac{2}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (x^2)^2 dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^4 dx$$

$$= \frac{2}{\pi} \left[\frac{x^5}{5} \right]_0^{\pi}$$

$$= \frac{2}{5\pi} [\pi^5 - 0]$$

$$= \frac{2\pi^4}{5}$$

$$\bar{y}^2 = \frac{2\pi^4}{5}$$

$$\bar{y} = \pi^2 \sqrt{\frac{2}{5}}$$

State Jordan's Lemma.